Equivalent Fractions and Mixed Numbers

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS numerator *(taurunga)* · denominator *(tauraro)* · equivalent fraction *(hautanga ōrite)*

A Equivalent fractions ●○○

Remember: Multiply the top and the bottom by the same number: $\frac{1}{2} = \frac{1 \times 3}{2 \times 3} = \frac{3}{6}$.

1. Fill in the box to make the fractions equal.

a) $\frac{1}{2} = \frac{\square}{4}$ **b)** $\frac{1}{3} = \frac{\square}{6}$ **c)** $\frac{1}{4} = \frac{\square}{12}$ **d)** $\frac{1}{2} = \frac{5}{\square}$

e) $\frac{2}{3} = \frac{\square}{6}$ **f)** $\frac{3}{4} = \frac{6}{\square}$ **g)** $\frac{1}{5} = \frac{\square}{10}$ **h)** $\frac{2}{5} = \frac{4}{\square}$

i) $\frac{3}{4} = \frac{\square}{12}$ **j)** $\frac{1}{10} = \frac{10}{\square}$ **k)** $\frac{2}{3} = \frac{\square}{12}$ **l)** $\frac{3}{5} = \frac{6}{\square}$

B Simplifying fractions ●●○

Remember: Divide the top and the bottom by the same number: $\frac{6}{8} = \frac{6 \div 2}{8 \div 2} = \frac{3}{4}$.

2. Write each fraction in its simplest form.

a) $\frac{2}{4}$ = _____ **b)** $\frac{3}{6}$ = _____ **c)** $\frac{2}{6}$ = _____ **d)** $\frac{4}{8}$ = _____

e) $\frac{6}{8}$ = _____ **f)** $\frac{5}{10}$ = _____ **g)** $\frac{4}{10}$ = _____ **h)** $\frac{3}{9}$ = _____

i) $\frac{6}{9}$ = _____ **j)** $\frac{8}{10}$ = _____ **k)** $\frac{9}{12}$ = _____ **l)** $\frac{10}{20}$ = _____

C How many parts? ●●○

3. Answer each question.

a) How many quarters are there in 1 whole? ____

b) How many halves are there in 2 wholes? ____

c) How many thirds are there in 3 wholes? ____

d) How many fifths are there in 2 wholes? ____

e) How many quarters are there in 4 wholes? ____

f) How many tenths are there in 3 wholes? ____

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D Improper fractions and mixed numbers ●●○

Remember: An improper fraction has a numerator larger than its denominator. $\frac{5}{4}$ is 5 quarters, which makes 1 whole and 1 quarter: $1\frac{1}{4}$.

4. Write each improper fraction as a mixed number.

a) $\frac{3}{2}$ = _____ **b)** $\frac{5}{4}$ = _____ **c)** $\frac{4}{3}$ = _____ **d)** $\frac{7}{5}$ = _____

e) $\frac{5}{2}$ = _____ **f)** $\frac{7}{3}$ = _____ **g)** $\frac{9}{4}$ = _____ **h)** $\frac{13}{10}$ = _____

5. Write each mixed number as an improper fraction.

a) $1\frac{1}{2}$ = _____ **b)** $1\frac{1}{4}$ = _____ **c)** $1\frac{2}{3}$ = _____ **d)** $2\frac{1}{2}$ = _____

e) $1\frac{3}{5}$ = _____ **f)** $2\frac{1}{3}$ = _____ **g)** $2\frac{3}{4}$ = _____ **h)** $1\frac{7}{10}$ = _____

E Comparing fractions ●●●

Remember: Change one fraction so that both have the same denominator, then compare the numerators.

6. Which fraction is larger?

a) $\frac{1}{2}$ or $\frac{1}{4}$ **b)** $\frac{1}{2}$ or $\frac{3}{4}$ **c)** $\frac{2}{3}$ or $\frac{5}{6}$

d) $\frac{3}{5}$ or $\frac{7}{10}$ **e)** $\frac{1}{4}$ or $\frac{3}{8}$ **f)** $\frac{1}{3}$ or $\frac{2}{9}$

7. Write each set of fractions in order, smallest first.

a) $\frac{1}{2}$, $\frac{1}{4}$, $\frac{3}{4}$ **b)** $\frac{1}{2}$, $\frac{3}{8}$, $\frac{5}{8}$ **c)** $\frac{2}{3}$, $\frac{1}{6}$, $\frac{1}{2}$

★ Extension

8. Write three different fractions that are equivalent to $\frac{1}{2}$.

9. A pizza is cut into 8 equal slices. 12 slices are eaten at a party. How many pizzas is that? Give your answer as a mixed number in its simplest form.

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Equivalent Fractions and Mixed Numbers

Name _____ Date _____ No calculator Version 2

Scan to practise this topic online

KEY WORDS numerator *(taurunga)* · denominator *(tauraro)* · equivalent fraction *(hautanga ōrite)*

A Equivalent fractions ●○○

Remember: Multiply the top and the bottom by the same number: $\frac{1}{2} = \frac{1 \times 3}{2 \times 3} = \frac{3}{6}$.

1. Fill in the box to make the fractions equal.

a) $\frac{1}{2} = \frac{\square}{6}$ **b)** $\frac{1}{4} = \frac{\square}{8}$ **c)** $\frac{1}{3} = \frac{\square}{9}$ **d)** $\frac{1}{2} = \frac{4}{\square}$

e) $\frac{3}{4} = \frac{\square}{8}$ **f)** $\frac{2}{3} = \frac{4}{\square}$ **g)** $\frac{1}{5} = \frac{\square}{15}$ **h)** $\frac{3}{5} = \frac{6}{\square}$

i) $\frac{2}{3} = \frac{\square}{9}$ **j)** $\frac{3}{10} = \frac{30}{\square}$ **k)** $\frac{3}{4} = \frac{\square}{16}$ **l)** $\frac{4}{5} = \frac{8}{\square}$

B Simplifying fractions ●●○

Remember: Divide the top and the bottom by the same number: $\frac{6}{8} = \frac{6 \div 2}{8 \div 2} = \frac{3}{4}$.

2. Write each fraction in its simplest form.

a) $\frac{2}{8}$ = _____ **b)** $\frac{3}{9}$ = _____ **c)** $\frac{2}{10}$ = _____ **d)** $\frac{4}{6}$ = _____

e) $\frac{6}{10}$ = _____ **f)** $\frac{5}{15}$ = _____ **g)** $\frac{4}{12}$ = _____ **h)** $\frac{2}{6}$ = _____

i) $\frac{8}{12}$ = _____ **j)** $\frac{6}{8}$ = _____ **k)** $\frac{10}{12}$ = _____ **l)** $\frac{20}{30}$ = _____

C How many parts? ●●○

3. Answer each question.

a) How many fifths are there in 1 whole? _____

b) How many halves are there in 3 wholes? _____

c) How many thirds are there in 2 wholes? _____

d) How many quarters are there in 2 wholes? _____

e) How many fifths are there in 3 wholes? _____

f) How many tenths are there in 2 wholes? _____

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D Improper fractions and mixed numbers ●●○

> **Remember:** An improper fraction has a numerator larger than its denominator. $\frac{5}{4}$ is 5 quarters, which makes 1 whole and 1 quarter: $1\frac{1}{4}$.

4. Write each improper fraction as a mixed number.

a) $\frac{5}{4}$ = _____ **b)** $\frac{3}{2}$ = _____ **c)** $\frac{5}{3}$ = _____ **d)** $\frac{6}{5}$ = _____

e) $\frac{7}{2}$ = _____ **f)** $\frac{8}{3}$ = _____ **g)** $\frac{11}{4}$ = _____ **h)** $\frac{17}{10}$ = _____

5. Write each mixed number as an improper fraction.

a) $1\frac{1}{3}$ = _____ **b)** $1\frac{3}{4}$ = _____ **c)** $1\frac{1}{5}$ = _____ **d)** $3\frac{1}{2}$ = _____

e) $1\frac{2}{5}$ = _____ **f)** $2\frac{2}{3}$ = _____ **g)** $2\frac{1}{4}$ = _____ **h)** $1\frac{3}{10}$ = _____

E Comparing fractions ●●●

> **Remember:** Change one fraction so that both have the same denominator, then compare the numerators.

6. Which fraction is larger?

a) $\frac{1}{2}$ or $\frac{1}{3}$ **b)** $\frac{1}{2}$ or $\frac{5}{8}$ **c)** $\frac{1}{3}$ or $\frac{1}{6}$

d) $\frac{2}{5}$ or $\frac{3}{10}$ **e)** $\frac{3}{4}$ or $\frac{5}{8}$ **f)** $\frac{2}{3}$ or $\frac{5}{9}$

7. Write each set of fractions in order, smallest first.

a) $\frac{3}{4}$, $\frac{1}{2}$, $\frac{1}{4}$ **b)** $\frac{1}{2}$, $\frac{7}{8}$, $\frac{3}{8}$ **c)** $\frac{1}{3}$, $\frac{5}{6}$, $\frac{1}{2}$

★ Extension

8. Write three different fractions that are equivalent to $\frac{1}{3}$.

9. A pizza is cut into 8 equal slices. 18 slices are eaten at a party. How many pizzas is that? Give your answer as a mixed number in its simplest form.

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